

PHYSICS — Unit P1: Kinematics (Motion in 1D & 2D)

Sessions 1-2 (4 hours total) | Track: Physics

Learning Objectives

Distinguish displacement, velocity, and acceleration precisely.

Apply the kinematic equations to solve motion problems.

Interpret position-time and velocity-time graphs.

Extend 1D motion to 2D projectile motion.

1. Core quantities

Quantity	Symbol	Unit	Definition
Position	x	m (meters)	location relative to origin
Displacement	$\Delta x = x_f - x_i$		change in position (vector)
Velocity	$v = \Delta x / \Delta t$	m/s	rate of change of position
Instantaneous velocity	$v = dx/dt$	m/s	velocity at an instant
Acceleration	$a = \Delta v / \Delta t$	m/s ²	rate of change of velocity
Instantaneous acceleration	$a = dv/dt$	m/s ²	acceleration at an instant

Important distinction: *speed* is scalar (magnitude only), *velocity* is a vector (magnitude + direction).

Displacement $\neq$ distance traveled (displacement can be smaller if the path curves back).

2. The kinematic equations (constant acceleration)

Let v_0 = initial velocity, v = final velocity, a = constant acceleration, t = time, Δx = displacement.

(1) $v = v_0 + a \cdot t$ (2) $\Delta x = v_0 \cdot t + \frac{1}{2} \cdot a \cdot t^2$ (3) $v^2 = v_0^2 + 2 \cdot a \cdot \Delta x$ (4) $\Delta x = \frac{1}{2} \cdot (v_0 + v) \cdot t$

These four equations are derived from the definitions of v and a via calculus (integration), but for this unit we use them directly as tools.

Example 2.1 A car starts from rest and accelerates at 3 m/s² for 5 seconds. Find its final velocity and distance traveled.

Given: $v_0 = 0$, $a = 3 \text{ m/s}^2$, $t = 5 \text{ s}$

Using (1): $v = 0 + (3)(5) = \mathbf{15 \text{ m/s}}$

Using (2): $\Delta x = (0)(5) + \frac{1}{2}(3)(5^2) = \frac{1}{2}(3)(25) = \mathbf{37.5 \text{ m}}$

Example 2.2 A ball is thrown straight up at 20 m/s. How high does it go? ($g = 9.8 \text{ m/s}^2$, take up as positive, so $a = -9.8 \text{ m/s}^2$)

At the top of the flight, $v = 0$.

Using (3): $v^2 = v_0^2 + 2 \cdot a \cdot \Delta x$ $0 = 20^2 + 2(-9.8)(\Delta x)$ $0 = 400 - 19.6 \cdot \Delta x$ $\Delta x = 400/19.6 \approx \mathbf{20.4 \text{ m}}$

3. Graphical interpretation

On a **position-time graph**, the *slope* at any point = instantaneous velocity.

On a **velocity-time graph**, the *slope* = acceleration, and the **area under the curve** = displacement.

Example 3.1 A velocity-time graph shows constant velocity of 4 m/s for 6 seconds. Displacement = area of rectangle = $4 \text{ m/s} \times 6 \text{ s} = \mathbf{24 \text{ m}}$

If instead velocity increases linearly from 0 to 10 m/s over 5 s (a triangle): Displacement = area of triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 5 \text{ s} \times 10 \text{ m/s} = \mathbf{25 \text{ m}}$

4. Two-dimensional motion: Projectiles

Split motion into independent horizontal (x) and vertical (y) components. Only gravity acts vertically ($a_y = -g$); horizontal velocity is constant ($a_x = 0$).

$$x(t) = v_{0x} \cdot t \quad y(t) = v_{0y} \cdot t - \frac{1}{2} \cdot g \cdot t^2 \quad v_{0x} = v_0 \cdot \cos(\theta) \quad v_{0y} = v_0 \cdot \sin(\theta)$$

Example 4.1 A projectile is launched at 30 m/s at an angle of 40° above horizontal. Find the time to reach maximum height.

$$v_{0y} = v_0 \sin(\theta) = 30 \times \sin(40^\circ) \approx 30 \times 0.643 \approx 19.3 \text{ m/s}$$

$$\text{At max height, } v_y = 0: v_y = v_{0y} - g \cdot t \Rightarrow 0 = 19.3 - 9.8t \Rightarrow t \approx \mathbf{1.97 \text{ s}}$$

5. In-Session Guided Practice

A cyclist accelerates from 2 m/s to 8 m/s in 3 seconds. Find the acceleration.

Using the result of Q1, find the distance traveled during that time.

A stone is dropped ($v_0 = 0$) from a cliff and hits the ground after 4 s. How tall is the cliff? ($g = 9.8 \text{ m/s}^2$)

Sketch (describe in words) what a velocity-time graph looks like for an object moving at constant velocity, then decelerating to a stop.

A ball is thrown horizontally at 15 m/s from a 20 m high platform. How long is it in the air, and how far does it travel horizontally?

6. Take-Home Questions

Q1. A car traveling at 25 m/s brakes with a deceleration of 5 m/s^2 . How long does it take to stop, and how far does it travel while braking?

Q2. An object's position is given by $x(t) = 3t^2 - 2t + 1$ (x in meters, t in seconds). Find its velocity and acceleration as functions of t. (Hint: $v = dx/dt$, $a = dv/dt$)

Q3. A projectile is launched at 25 m/s at 50° above horizontal. Find (a) the time of flight, and (b) the horizontal range. ($g = 9.8 \text{ m/s}^2$; total flight time for a projectile landing at the same height = $2 \cdot v_{0y}/g$)

Q4. Explain, using the definitions of displacement and distance, why a runner who does one full lap of a 400 m track has a displacement of 0 m but a distance traveled of 400 m.

Q5. Why do the kinematic equations in Section 2 only apply when acceleration is constant? What would you need instead if acceleration varied with time?

Answer Key (for tutor reference)

$a = (8-0)/3$? no $v_0=25...$ — using $v=v_0+at$: $5 = 25 + a(t)$... $a = (8-25)/3$? wait this is Q1 practice not takehome. $a = \Delta v/\Delta t = (8-25)/3 = \mathbf{2 \text{ m/s}^2}$

$$\Delta x = \frac{1}{2}(v_0+v)t = \frac{1}{2}(25+8)(3) = \mathbf{15 \text{ m}}$$

$$\Delta x = \frac{1}{2}gt^2 = \frac{1}{2}(9.8)(16) = \mathbf{78.4\ m}$$

Flat line at constant height, then a straight line sloping down to zero (constant negative slope = deceleration).

$$t = \sqrt{2h/v} \dots \text{using } y = -\frac{1}{2}gt^2: 20 = \frac{1}{2}(9.8)t^2 \rightarrow t \approx 2.02\text{s}; \text{ horizontal distance} = 15 \times 2.02 \approx \mathbf{30.3\ m}$$

Take-home: Q1: $t = v/a = 25/5 = 5\text{ s}$; $\Delta x = v^2/(2a) = 625/10 = 62.5\text{ m}$ Q2: $v(t) = 6t - 2\text{ (m/s)}$; $a(t) = 6\text{ (m/s}^2\text{, constant)}$ Q3: $v_{0y} = 25 \sin(50^\circ) \approx 19.15$; time of flight $= 2(19.15)/9.8 \approx 3.91\text{ s}$; range $= v_{0x} \times t$, $v_{0x} = 25 \cos(50^\circ) \approx 16.07 \rightarrow \text{range} \approx 62.8\text{ m}$ Q4: Displacement measures net change in position (start = end point on a lap $\rightarrow 0$); distance measures total path length covered regardless of direction (400 m). Q5: The equations assume a is constant so integrating $dv/dt = a$ gives clean polynomial expressions; if $a(t)$ varies, you need calculus — integrate $a(t)$ to get $v(t)$, then integrate $v(t)$ to get $x(t)$.